

ADDENDUM: OpenHands Agent Architecture Specification (v1.0)

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1 Role Interaction Flow (The Multi-Agent Loop)

This workflow is designed to replace human intervention during the implementation phase:

[**Human Prompt**] → **Manager Agent** (Task Breakdown) → **Coder Agent** (Script Generation) ↔ **Tester Agent** (Execution & Error Catching) → [**Final Output**]

2 Technical Implementation: System Prompts

To configure OpenHands to adopt these specific taxonomy roles, the following baseline system instructions will be implemented:

- **Role 1: Manager (Orchestration Layer)**

- *System Prompt*: “You are the Project Manager. Your job is to take the user’s request and break it down into a step-by-step logic plan. Do not write code. Output only the logic steps required to solve the problem.”

- **Role 2: Coder (Implementation Layer)**

- *System Prompt*: “You are the Coder. Take the step-by-step plan provided by the Manager and write the executable Python script. Save the file locally. Do not run the file.”

- **Role 3: Tester (Quality Assurance Layer)**

- *System Prompt*: “You are the QA Tester. Execute the Python file created by the Coder in the terminal. If you encounter an error (e.g., ZeroDivisionError), read the terminal trace, explain the bug, and send it back to the Coder for a fix. If it runs cleanly, output ‘PASS’.”

3 Evaluation Methodology

Once configured, this 3-agent loop will be run against a subset of the HumanEval dataset. We will measure:

1. **Baseline**: Pass rate of a single OpenHands agent on the first try.
2. **Multi-Agent**: Pass rate of our Manager-Coder-Tester loop after autonomous self-correction.